1. Pointer Manipulation — Referencing and Dereferencing

Write a Go program demonstrating pointer usage:

(1) declare a variable, print its address using & and access its value using *;

(2) write a function that accepts a pointer parameter and modifies the original variable (pass-by-reference) — print before and after values;

(3) allocate a struct using new() and access/modify its fields through the pointer.



```go
package main

import "fmt"

// Function to modify original variable using pointer
func modifyValue(ptr *int) {
    *ptr = 100
}

func main() {

    // Part 1: Pointer Basics
    var num int = 50

    fmt.Println("----- POINTER BASICS -----")

    fmt.Println("Original Value:", num)
    fmt.Println("Memory Address:", &num)
    fmt.Println("Value using Pointer:", *(&num))

    // Part 2: Modify variable using pointer function
    fmt.Println("\n----- POINTER FUNCTION -----")

    fmt.Println("Before Modification:", num)

    modifyValue(&num)

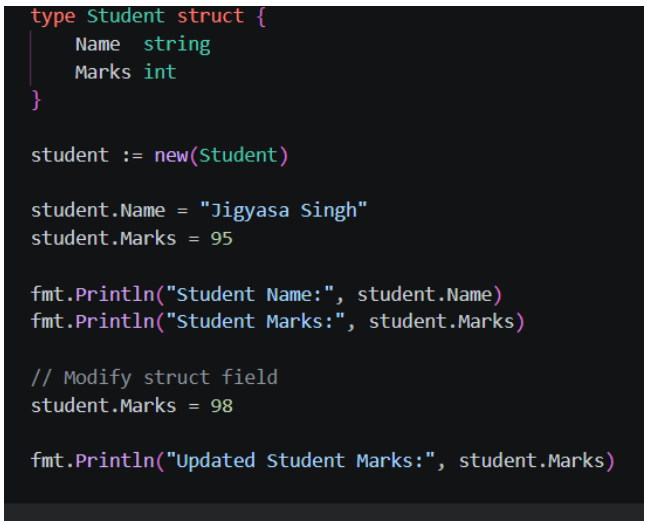
    fmt.Println("After Modification:", num)

    // Part 3: Struct Pointer using new()
    fmt.Println("\n----- STRUCT POINTER -----")

    type Student struct {
        Name  string
        Marks int
    }

    student := new(Student)

    student.Name = "Jigyasa Singh"
    student.Marks = 95
```



```go
type Student struct {
    Name  string
    Marks int
}

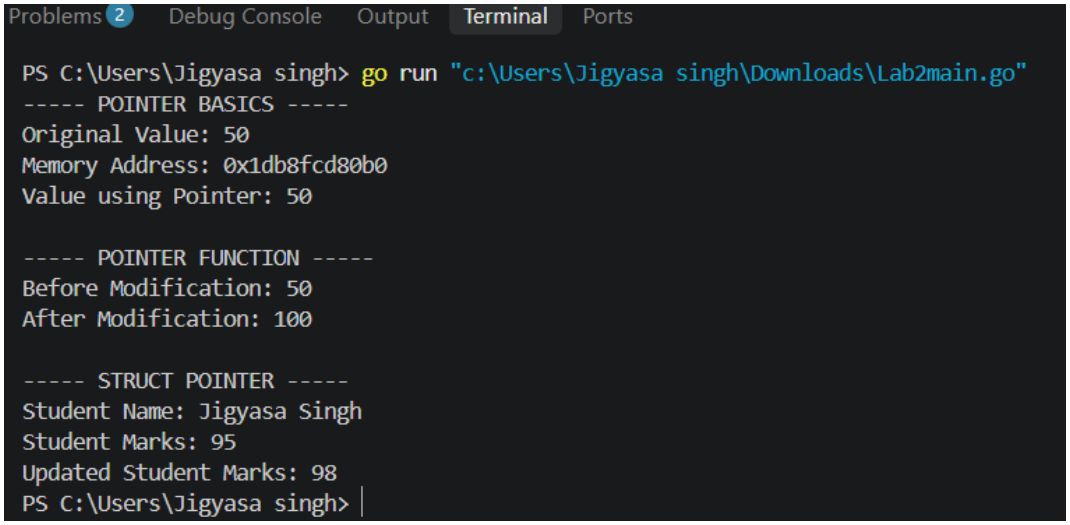
student := new(Student)

student.Name = "Jigyasa Singh"
student.Marks = 95

fmt.Println("Student Name:", student.Name)
fmt.Println("Student Marks:", student.Marks)

// Modify struct field
student.Marks = 98

fmt.Println("Updated Student Marks:", student.Marks)
```


Problems 2
Debug Console
Output
Terminal
Ports
PS C:\Users\Jigyasa singh> go run "c:\Users\Jigyasa singh\Downloads\Lab2main.go"
----- POINTER BASICS -----
Original Value: 50
Memory Address: 0x1db8fcd80b0
Value using Pointer: 50
----- POINTER FUNCTION -----
Before Modification: 50
After Modification: 100
----- STRUCT POINTER -----
Student Name: Jigyasa Singh
Student Marks: 95
Updated Student Marks: 98
PS C:\Users\Jigyasa singh>